

Team Number: _____

Overall Project Performance (15 marks in total)

Criteria (3 marks each)	0 mark	1-2 mark	3 marks
Fall detection reliability (True Positives)	Rarely detects a fall, or detection appears random/unrelated to the fall motion.	Detects a fall sometimes, but requires repeated attempts or only works for very obvious falls.	Reliably detects falls across repeated trials and small variations in height, angle, or motion.
False alarms during normal wearable such as walking use (False Positives)	Frequently reports a fall during normal actions such as carrying, walking-like motion, tilting, or light shaking.	Occasional false alarms occur under certain normal movements.	Rare or near-zero false alarms occur during typical non-fall use. Ei. Light shaking, casual tilting.
Distinguishing near-fall movements such as sitting down	Often misclassifies sitting, slow bending, slow tilting, or controlled lowering as falls.	Distinguishes some near-fall actions but still misclassifies common examples.	Consistently separates common near-fall actions from real falls. Ei. Medium speed bending, slow-controlled lowering.
Criteria (2 marks each)	0 mark	1 mark	2 marks
Wearable system architecture and data flow	No clear architecture is presented, or the described data flow does not match the implemented system. Major stages are missing. The assembly EWMA is not meaningfully integrated, is applied only to part of the motion data, uses incorrect/shared recursive state, or the detection logic bypasses the filtered outputs.	A generally correct architecture is presented, but some links between sensing, filtering, detection logic, states, or outputs are incomplete or unclear. EWMA is integrated for accelerometer and gyroscope data, but there are minor issues in axis coverage, state handling, alpha use, or consistency of filtered-data usage.	A clear and accurate architecture shows sensor acquisition -> assembly EWMA -> feature extraction -> fall-detection/state logic -> outputs/alerts, and matches the implemented code. The assembly EWMA is correctly applied to all required accelerometer and gyroscope axes with separate recursive states, and the filtered outputs are consistently used by the detection pipeline.
Sensor fusion, feature selection, and thresholding quality	Uses only one motion sensor, or features/thresholds are arbitrary, poorly interpreted, or not meaningfully related to the physical motion.	Uses both accelerometer and gyroscope data, but feature selection, units, thresholds, or justification are only partly effective.	Uses accelerometer and gyroscope data coherently with practical, explainable features and thresholds in appropriate physical units.
Responsiveness, state stability, and alert behaviour	Large delay, unstable/flickering classification, or LED/alert behaviour does not match the system state.	Reasonable response but with brief instability, weak confirmation logic, or occasionally inconsistent LED/alert timing.	Prompt and stable state transitions; normal/fall indications are consistent, with effective confirmation, debounce, or hysteresis where appropriate.

Enhancement and Additional Elderly-Wearable Features (9 marks in total)

Criteria	0 mark	1 mark	3 marks
Relevance, usefulness and innovative to an elderly user	Enhancement is missing or cosmetic and does not improve safety, usability, independence, or response.	Some relevance, but limited practical impact on the wearable.	Clearly improves fall reliability and/or adds a meaningful elderly-support function such as Long Lie escalation, SOS, inactivity monitoring, alerting, or logging.
Technical depth	Very minimal change with little technical challenge, such as trivial prints or a cosmetic LED change.	Moderate technical work; useful logic is added but sophistication or evaluation is limited.	Non-trivial implementation such as a state-machine extension, adaptive filtering, sensor fusion, communication, logging, or a meaningful algorithmic change.
Integration quality and robustness	Breaks the core system, is unstable, or interferes with filtering, detection, UART, or LED behaviour.	Works but is fragile, has edge-case glitches, or noticeably degrades core performance.	Well integrated and robust; does not degrade compulsory fall detection and remains stable throughout the demonstration.

Penalty Deduction

Criteria	Item Description	Marks to Deduct
Code import and debug configuration	Takes more than 5 minutes to compile/run, or requires GA intervention to import or start the project.	-2
Overall implementation	Glitched demo, unstable state, poorly tuned threshold, or system restart required.	-1 per issue, max -3
Code quality	Bad coding style, unclear structure, messy comments, or poor variable/function names.	-1
Assembly compliance	C helper used for the compulsory EWMA routine in the application, or callee-saved registers are not preserved.	-2

Presentation student A, Name: _____ (4 marks in total, individual contribution)

Criteria (1 mark each)	0 mark	0.5 mark	1 mark
Content and clarity	The wearable, EWMA filter, fall logic, or modes are unclear or missing. Sensor roles and state transitions are not explained.	Some explanation is provided, but important technical details, sensor roles, alpha/threshold choices, or state transitions are incomplete.	The wearable's purpose, EWMA assembly, sensor processing, fall state machine, testing evidence, and enhancements are clear, thorough, accurate, and engaging.
Organization and flow	No clear structure; topics jump between code, hardware, results, and features without guiding the audience.	Some structure is apparent, but transitions are weak or sections are disorganized.	Logically structured with smooth transitions and a clear narrative from elderly-user need to technical design, evidence, and outcome.
Communication skills	The student struggles to explain technical aspects; limited eye contact and ineffective delivery make the presentation difficult to follow.	Some effort to explain technical concepts, but engagement, confidence, or clarity is inconsistent.	Technical details are explained confidently and accessibly; tone, pacing, body language, and audience engagement are effective throughout.
Live demonstration integration	Demonstration is missing, unclear, unsafe, or disconnected from the explanation.	A demonstration is present but incomplete, weakly connected, or does not clearly show the required wearable behaviours.	A safe, well-rehearsed demonstration is fully integrated and clearly shows filtering, normal activity, near-fall discrimination, fall response, stability, and enhancements.

Q&A, Student A: _____ (2 marks in total)

Presentation student B, Name: _____ (4 marks in total, individual contribution)

Criteria (1 mark each)	0 mark	0.5 mark	1 mark
Content and clarity	The wearable, EWMA filter, fall logic, or modes are unclear or missing. Sensor roles and state transitions are not explained.	Some explanation is provided, but important technical details, sensor roles, alpha/threshold choices, or state transitions are incomplete.	The wearable's purpose, EWMA assembly, sensor processing, fall state machine, testing evidence, and enhancements are clear, thorough, accurate, and engaging.
Organization and flow	No clear structure; topics jump between code, hardware, results, and features without guiding the audience.	Some structure is apparent, but transitions are weak or sections are disorganized.	Logically structured with smooth transitions and a clear narrative from elderly-user need to technical design, evidence, and outcome.
Communication skills	The student struggles to explain technical aspects; limited eye contact and ineffective delivery make the presentation difficult to follow.	Some effort to explain technical concepts, but engagement, confidence, or clarity is inconsistent.	Technical details are explained confidently and accessibly; tone, pacing, body language, and audience engagement are effective throughout.
Live demonstration integration	Demonstration is missing, unclear, unsafe, or disconnected from the explanation.	A demonstration is present but incomplete, weakly connected, or does not clearly show the required wearable behaviours.	A safe, well-rehearsed demonstration is fully integrated and clearly shows filtering, normal activity, near-fall discrimination, fall response, stability, and enhancements.

Q&A, Student B: _____ (2 marks in total)

Slides (4 marks in total)

Criteria (1 mark each)	0 mark	0.5 mark	1 mark
Content accuracy	Key information is missing or technically inaccurate, especially the EWMA equation/interface, sensor units, state logic, or results.	Some key details are provided, but there are gaps in accuracy or completeness.	All important technical details are accurately presented, including the assembly interface, recursive state, sensor processing, detection logic, evidence, and limitations.
Visual design	Slides are cluttered, text-heavy, difficult to read, or visually inconsistent.	Slides are somewhat clear but have readability, contrast, density, or consistency issues.	Slides are visually appealing, professional, consistent, and easy to read; the design improves understanding and engagement.
Use of diagrams, graphics, and evidence	No useful diagrams/flowcharts/plots, or graphics are inappropriate and unsupported.	Some diagrams or plots are used but may be difficult to interpret, weakly labelled, or disconnected from claims.	Well-chosen and clearly labelled pipeline/state diagrams, test evidence, and visuals directly support the technical explanation and conclusions.
Slide structure and flow	Slides lack organization and present ideas randomly without guiding the audience.	Some organization is attempted, but ideas are disjointed or transitions are unclear.	Slides follow a logical story: user problem, requirements, filter design, implementation, detection logic, evidence, enhancements, and conclusions.